Simple Linear Regression

STAT 223 – Applied Analytics

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Simple Linear Regression

A **simple linear regression** is a way to model the linear relationship between two quantitative variables using a line drawn known as a regression line.

The **population regression** is defined as the equation:

$$\mathbb{E}\left[\, Y \,\right] = \beta_0 + \beta_1 X,$$

where $\mathbb{E}\left[\, Y \,\right]$ is the expected value of Y, given that value of X.

Simple Linear Regression

The **linear regression model** is defined as:

$$\widehat{Y} = \beta_0 + \beta_1 X + \varepsilon,$$

where $\widehat{Y}$ is the predicted value of Y, given that value of X and $\varepsilon \sim \textsc{Normal}(0, \sigma_e^2)$, where σ_e^2 is a constant variance across all values of X.

Note that, as $\varepsilon \sim \textsc{Normal}(0,\ \sigma_e^2)$ and β_0, β_1 and X are "fixed", then $\widehat{Y}$ is a random variable and $\widehat{Y} \sim \textsc{Normal}(\beta_0 + \beta_1 X,\ \sigma_e^2)$.

Least Squares Regression

Obviously, we rarely, if ever, know the values for the entire population, so we can't ever truly know β_0 and β_1 directly.

We have to estimate them from the sample values.

The **method of least squares** derives a linear regression model by minimizing the sum of squared errors:

$$\sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i)^2$$

The Theory

The first step in minimization is to determine the objective (target) function that needs to be minimized.

$$
\begin{aligned}
Q &= \sum_i \varepsilon_i^2 \\
&= \sum_i (Y_i - \hat{Y}_i)^2 \\
&= \sum_i (Y_i - (\beta_0 + \beta_1 X_i))^2
\end{aligned}
$$

This is our objective function. To minimize it, we take its derivative with respect to each parameter, set each equal to 0, and solve for the parameter.

The Theory

$$\begin{aligned}\frac{\partial}{\partial \beta_0} Q &= \frac{\partial}{\partial \beta_0} \sum_i (Y_i - \beta_0 - \beta_1 X_i)^2 \\ &= \sum_i 2(Y_i - \beta_0 - \beta_1 X_i)(-1)\end{aligned}$$

$$\begin{aligned}0 &\stackrel{\text{set}}{=} -2\left(\sum_i Y_i - \sum_i b_0 - \sum_i b_1 X_i\right) \\ &= n\overline{Y} - nb_0 - n\overline{X}b_1\end{aligned}$$

$$\implies \qquad b_0 = \overline{Y} - b_1 \overline{X}$$

Do not forget the definition of the sample mean:

$$\overline{X} = \frac{1}{n}\sum X_i \iff \sum X_i = n\overline{X}$$

The Theory

$$\frac{\partial}{\partial \beta_1} Q = \frac{\partial}{\partial \beta_1} \sum_i (Y_i - \beta_0 - \beta_1 X_i)^2$$
$$= \sum_i 2(Y_i - \beta_0 - \beta_1 X_i)(-X_i)$$

$$0 \stackrel{\text{set}}{=} -2 \left(\sum_i X_i Y_i - \sum_i b_0 X_i - \sum_i b_1 X_i \right)$$
$$= \sum_i X_i Y_i - (\overline{Y} - b_1 \overline{X}) \sum_i X_i - \sum_i b_1 X_i^2$$
$$= \sum_i X_i Y_i - (\overline{Y} - b_1 \overline{X}) n \overline{X} - b_1 \sum_i X_i^2$$
$$= \sum_i X_i Y_i - n \overline{XY} - b_1 n \overline{X}^2) - b_1 \sum_i X_i^2$$

The Theory

$$0 = \sum_i X_i Y_i - n\overline{XY} - b_1 n\overline{X}^2) - b_1 \sum_i X_i^2$$

$$b_1 \sum_i X_i^2 - b_1 n\overline{X}^2 = \sum_i X_i Y_i - n\overline{XY}$$

$$b_1 = \frac{\sum X_i Y_i - n\overline{XY}}{\sum X_i^2 - n\overline{X}^2}$$

The Theory

Thus, our estimators of the y-intercept and slope (effect) are:

$$b_0 = \overline{Y} - b_1\overline{X}$$
$$b_1 = \frac{\sum X_i Y_i - n\overline{XY}}{\sum X_i^2 - n\overline{X}^2}$$

The second formula can also be written as:

$$b_1 = \frac{\sum (X_i - \overline{X})(Y_i - \overline{Y})}{\sum (X_i - \overline{X})^2}$$
$$= r\left(\frac{s_Y}{s_X}\right)$$

This gives some insight into the slope, the correlation, and the relationship between them.

Formulas for Linear Regression

To estimate the line of best fit, we have the formula below:

$$\widehat{Y} = b_0 + b_1 X$$

And we can calculate b_0 and b_1 with the formulas:

$$b_1 = r\left(\frac{s_Y}{s_X}\right)$$
$$b_0 = \overline{Y} - b_1\overline{X}$$

Remember, we have to find the slope, b_1, first before we can find the y-intercept, b_0.

Residuals

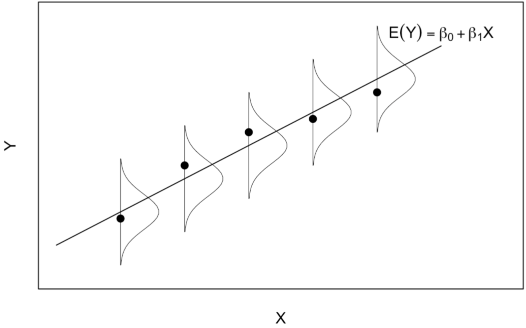
The difference between Y_i, the *actual* value, and $\widehat{Y}_i = b_0 + b_1 X_i$, the predicted value of Y given X_i, is called the **residual**.

$$\varepsilon_i = Y_i - \widehat{Y}_i$$

Least Squares Regression

These residuals must satisfy:

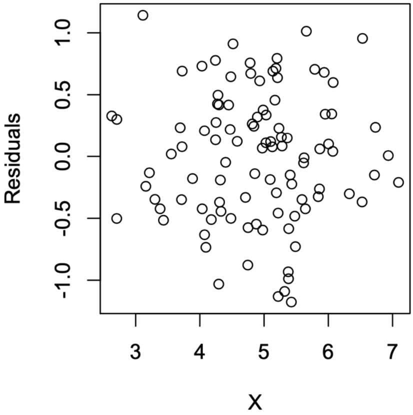
1. $\mathbb{E}[\varepsilon_i] = 0$.
2. σ_e^2 is the same for all x.
3. ε_i follows a NORMAL distribution.
4. $\varepsilon_i \perp \varepsilon_j$ for $i \neq j$.

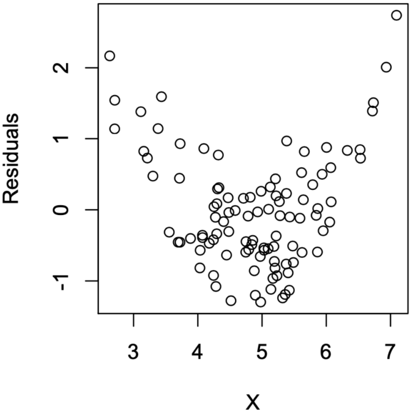


Residual Plots

We can check these assumptions with residual plots.

To check if the mean is zero:

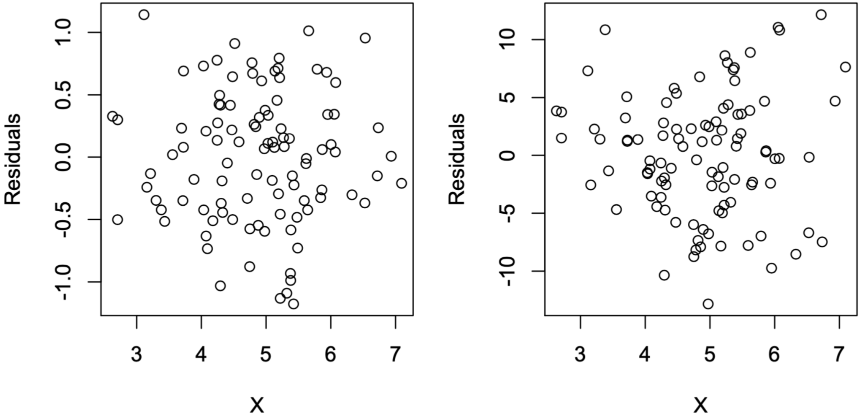




Residual Plots

We can check these assumptions with residual plots.

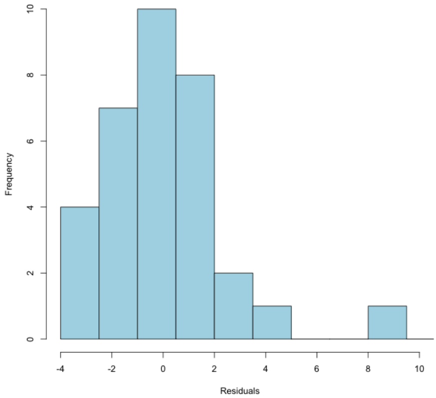
To check if the variance is constant:



Residual Plots

We can check these assumptions with residual plots.

To check if the residuals are normally distributed, you can check a histogram of the residuals:

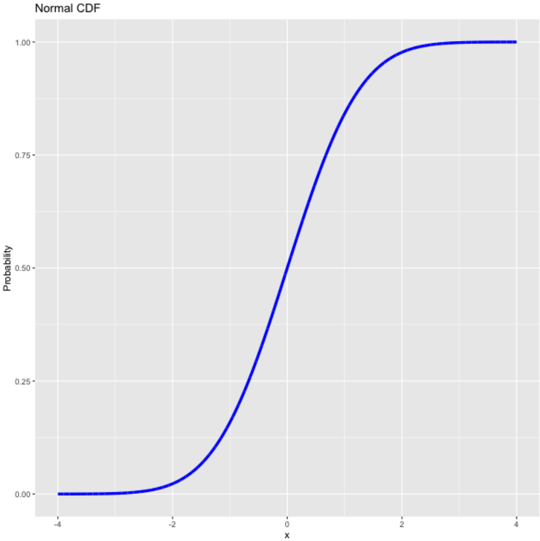


Residual Plots

To check if the residuals are normally distributed, you can run a Shapiro-Wilk test. (`shapiro` in the `scipy.stats` package in Python) We can also look at a Q-Q plot.

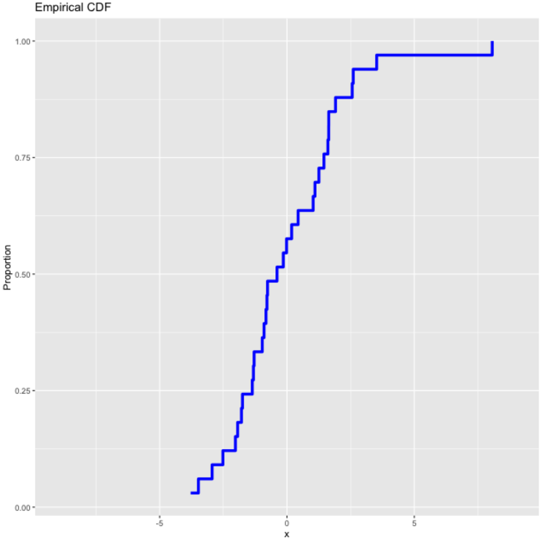
Q-Q Plot

Recall the cumulative distribution function:



Q-Q Plot

We can also create an *empirical cumulative distribution function*:



Q-Q Plot

The idea is to compare them, comparing the theoretical quantiles to the observed ones of the residuals. If the residuals fit along the line $y = x$, then we can say the distribution is NORMAL.